

SHOPEZ

E-Commerce Application
Final Project Report

Full Stack Development with MERN

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1. Introduction

1.1 Project Overview

SHOPEZ is a full-stack E-commerce web application developed using the MERN Stack (MongoDB, Express.js, React.js, and Node.js). The platform gives customers a seamless online shopping experience through a responsive interface, secure authentication, efficient product browsing, shopping cart management, and order processing. Administrators are provided with a dedicated dashboard to manage products, categories, users, inventory, and customer orders.

1.2 Purpose

The purpose of SHOPEZ is to provide a modern, secure, and user-friendly online shopping platform that simplifies how customers discover products, manage their cart, and complete purchases — while giving administrators the tools to manage the storefront efficiently. The project also serves as a practical implementation of MERN Stack technologies, giving the development team hands-on experience with frontend development, REST API design, database management, authentication, and deployment.

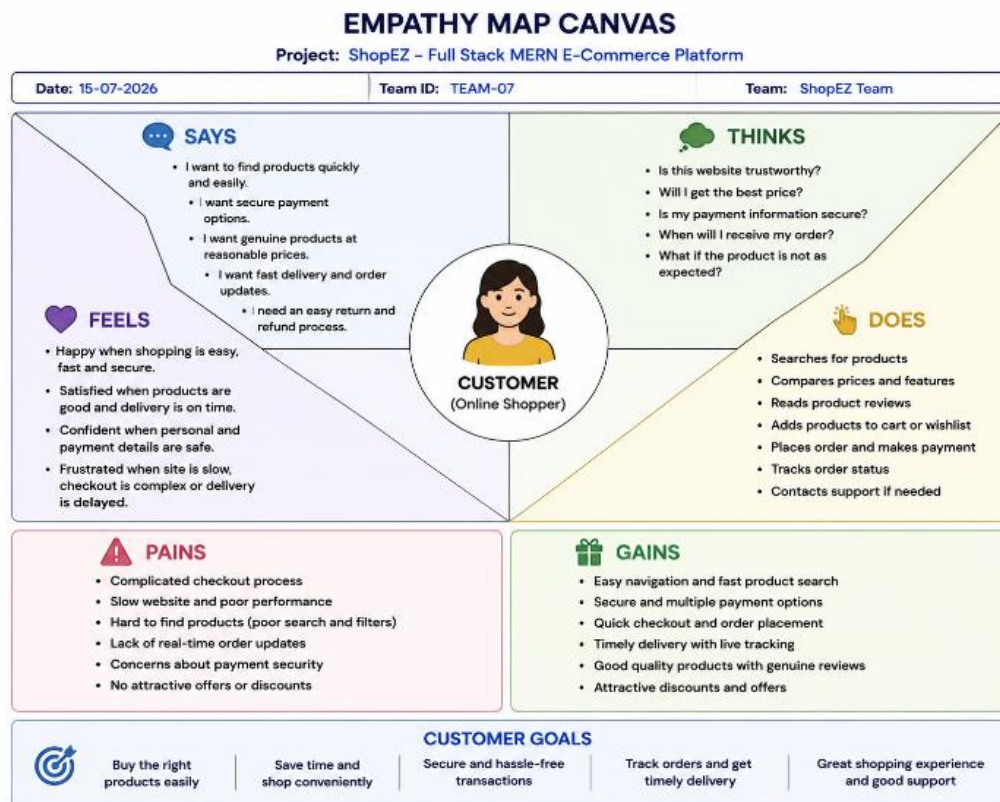
2. Ideation Phase

2.1 Problem Statement

Many small and mid-sized sellers lack access to an affordable, easy-to-manage online storefront, while customers often struggle with cluttered interfaces, unreliable product search, and unclear order tracking on existing platforms. There is a need for a lightweight, responsive e-commerce solution that makes it simple for customers to browse, purchase, and track orders, and for administrators to manage products and orders without a steep learning curve.

2.2 Empathy Map Canvas

The empathy map below captures what target users say, think, do, and feel when shopping online, and was used to guide the feature set for SHOPEZ.



2.3 Brainstorming

Key ideas generated during brainstorming sessions, later prioritized into the SHOPEZ feature set:

- A single, responsive storefront that works across desktop and mobile.
- Category-based browsing with search and filtering to reduce time-to-find.
- A persistent shopping cart with quantity controls and a running total.
- Secure, token-based authentication so users don't need to re-login often.
- An order history and status tracker so customers know where their order stands.
- A dedicated admin dashboard to manage products, users, and orders in one place.
- Room to add payments, reviews, and recommendations in later iterations.

Ideation Phase

Brainstorm & Idea Prioritization Template

Date: 15-07-2026	Team ID: TEAM-07	Project Name: ShopEZ – Full Stack MERN E-Commerce Platform	Maximum Marks: 4 Marks
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 Step-1: Team Gathering, Collaboration and Select the Problem Statement

Problem Identified	Reason
Existing e-commerce platforms are often complex, cluttered, and difficult to manage for both customers and administrators.	Users need a simple, secure, responsive, and efficient shopping platform with easy navigation and centralized admin management.

 Step-2: Brainstorm, Idea Listing and Grouping

Idea ID	Idea	Category
I-1	Responsive modern UI	User Experience
I-2	Secure JWT Authentication	Security
I-3	Product Search & Filters	Product Browsing
I-4	Shopping Cart	Shopping
I-5	Wishlist	Shopping
I-6	Coupon & Discount System	Sales
I-7	Razorpay Payment Integration	Payment
I-8	Order Tracking	Customer Service
I-9	Admin Dashboard	Administration
I-10	Product Management	Administration
I-11	Category Management	Administration
I-12	Banner Management	Administration
I-13	Dashboard Analytics	Reports
I-14	Customer Reviews	Product Feedback
I-15	Cloudinary Image Upload	Media Storage

 Step-3: Idea Prioritization

Idea	Business Value	Implementation Effort	Priority
User Authentication	High	Medium	High
Product Catalog	High	Medium	High
Shopping Cart	High	Medium	High
Checkout & Orders	High	High	High
Admin Dashboard	High	High	High
Product Management	High	Medium	High
Search & Filters	High	Low	High
Wishlist	Medium	Low	Medium
Coupons	Medium	Medium	Medium
Dashboard Analytics	Medium	Medium	Medium
Customer Reviews	Medium	Medium	Medium
Banner Management	Low	Low	Low

 Note: Ideas with High Business Value and Low/Medium Implementation Effort are considered top priority.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	Customer Actions	Touchpoints	Opportunities
Awareness	Discovers SHOPEZ via search or referral	Landing page, Home page	Highlight featured products and offers
Consideration	Browses categories, searches, compares products	Product listing, Product details	Fast search, clear filtering
Decision	Adds items to cart, reviews cart summary	Shopping cart	Simple quantity edits, visible totals
Purchase	Checks out and places order	Checkout page	Minimal steps, clear confirmation
Retention	Tracks order, returns to buy again	Order history, Profile	Order status updates, easy reordering

3.2 Solution Requirement

Functional Requirements

- User registration and JWT-based login
- Product listing, search, and category filtering
- Shopping cart with add/update/remove operations
- Order placement, history, and status tracking
- Admin dashboard for product, user, and order management

Non-Functional Requirements

- Responsive UI across desktop, tablet, and mobile
- Secure password hashing and token validation
- Reasonable API response times under normal load
- Input validation and consistent error handling
- Maintainable, modular codebase (MVC-style backend, componentized frontend)

3.3 Data Flow Diagram

The diagram below illustrates how data moves between the user, the React frontend, the Express/Node backend, and the MongoDB database.



3.4 Technology Stack

Layer	Technology	Purpose
Frontend	React.js, React Router, Axios	UI rendering, client-side routing, API calls
Backend	Node.js, Express.js	REST API, routing, middleware, business logic
Database	MongoDB, Mongoose	Data storage and schema modeling
Authentication	JSON Web Tokens (JWT), bcrypt	Secure login sessions and password hashing
Tooling	Git & GitHub, Postman, VS Code	Version control, API testing, development

4. Project Design

4.1 Problem – Solution Fit

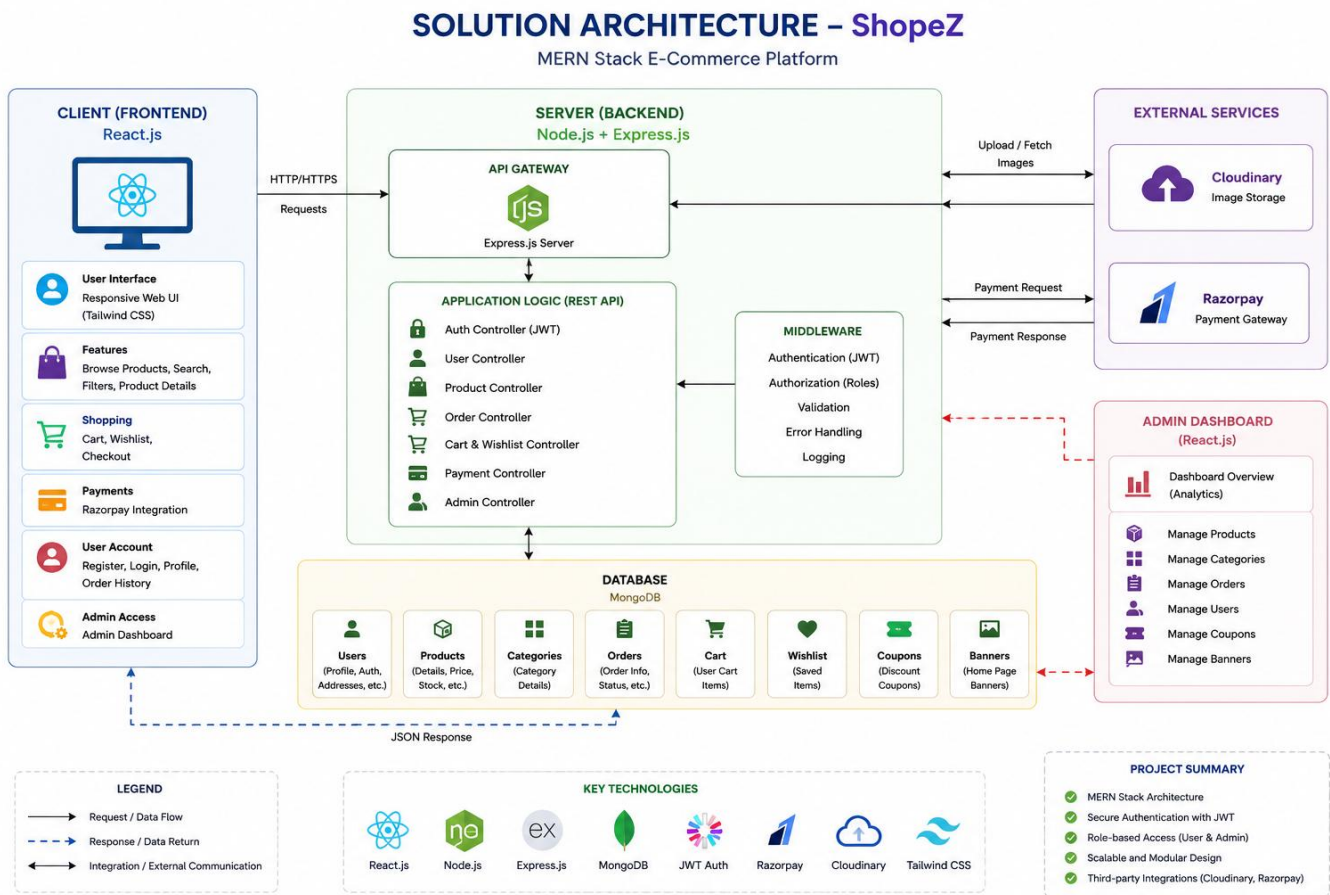
Customers need a fast, uncluttered way to browse and buy products online, and administrators need a simple way to manage inventory and orders. SHOPEZ addresses this by pairing a lightweight React storefront with a role-based admin dashboard, backed by a single REST API — so both audiences get a purpose-built interface without added complexity.

4.2 Proposed Solution

SHOPEZ is proposed as a three-tier MERN application: a React.js frontend for browsing, cart, and checkout; an Express/Node.js backend exposing authenticated REST APIs; and a MongoDB database for products, users, carts, and orders. JWT-based authentication secures both customer and admin routes, with role checks gating administrative operations.

4.3 Solution Architecture

The architecture follows a client-server model with clear separation between presentation, application, and data layers.



5. Project Planning & Scheduling

5.1 Project Planning

The project was planned using an agile sprint-based approach. Epics were broken into user stories, estimated in story points, and scheduled across sprints. A summary of the sprint schedule is below — see the full Project Planning document for the complete backlog, tracker, and burndown details.

Sprint	Duration	Story Points Planned	Story Points Completed	Release Date
Sprint-1	[]	[]	[]	[]
Sprint-2	[]	[]	[]	[]
Sprint-3	[]	[]	[]	[]
Sprint-4	[]	[]	[]	[]

6. Functional and Performance Testing

Functional testing covered user registration, login, product browsing, cart operations, order placement, profile management, and admin dashboard workflows, using manual testing and Postman for API verification.

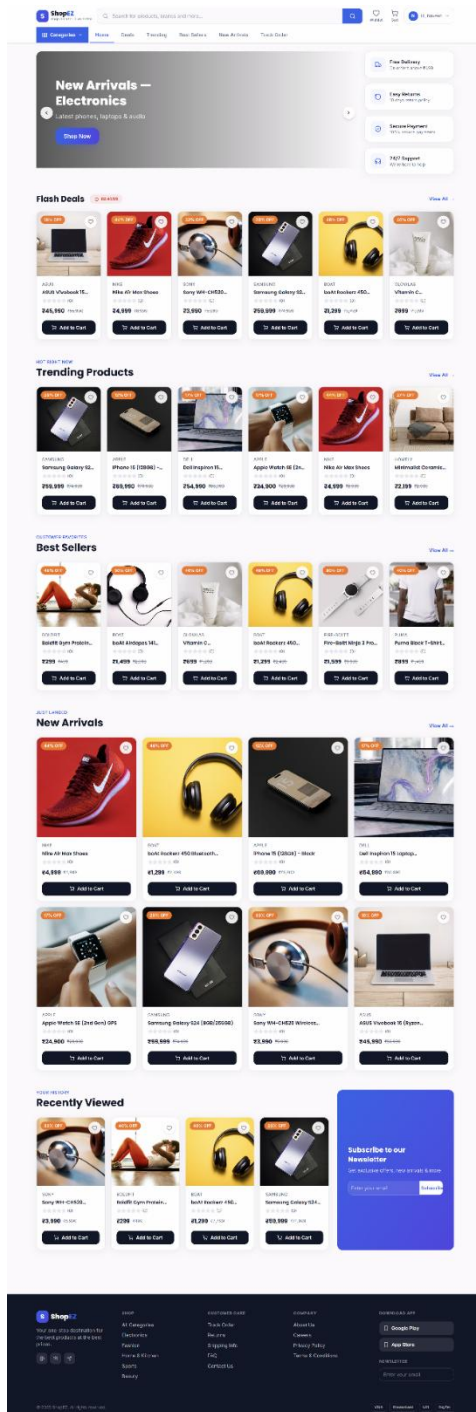
6.1 Performance Testing

Metric	Target	Observed Result
Average API response time	[]	[]
Page load time (Home page)	[]	[]
Concurrent users supported	[]	[]
Login success rate	[]	[]

7. Results

7.1 Output Screenshots

Home Page



Login / Registration

S

Welcome back

Sign in to continue to ShopEZ

[Forgot password?](#)

Sign In

[Don't have an account? Create one](#)

S

Create your account

Join ShopEZ for a premium shopping experience

Create Account

[Already have an account? Sign in](#)

Product Listing

S

ShopEZ

Shop Smarter. Live Better.

Wishlist

Cart

Hi, Naveen

Categories

Home

Deals

Trending

Best Sellers

New Arrivals

Track Order

Home

All Products

Filters

Category

☒ All

☐ Beauty

☐ Books

☐ Electronics

☐ Fashion

☐ Home & Kitchen

☐ Sports

☐ Toys

Brand

☐ ASUS

☐ Apple

☐ Boldfit

☐ Dell

☐ Fire-Bolt

☐ GlowLab

☐ Homely

☐ Lenovo

[View More](#)

Price

to

Rating

☐ 4* & above

☐ 3* & above

Showing 1-12 of 16 products

Sort by: Popularity

27% OFF



HOMELY

Minimalist Ceramic Vase...

★★★★★ (0)

₹2,199 ₹2,999

Add to Cart

40% OFF



BOLDFIT

Boldfit Gym Protein Shak...

★★★★★ (0)

₹299 ₹499

Add to Cart

48% OFF



BOAT

boAt Rockerz 450...

★★★★★ (0)

₹1,299 ₹2,499

Add to Cart

44% OFF



NIKE

Nike Air Max Shoes

★★★★★ (0)

₹4,999 ₹8,999

Add to Cart

30% OFF



REDMI

Redmi 13C 6GB/128GB...

★★★★★ (0)

₹8,999 ₹12,999

Add to Cart

17% OFF



DELL

Dell Inspiron 15 Laptop...

★★★★★ (0)

₹54,990 ₹65,990

Add to Cart

48% OFF



GLOWLAB

Vitamin C Brightening...

★★★★★ (0)

₹699 ₹1,299

Add to Cart

12% OFF



APPLE

iPhone 15 (128GB) - Black

★★★★★ (0)

₹69,990 ₹79,900

Add to Cart

Shopping Cart

S

ShopEZ

Shop Smarter. Live Better.

Search for products, brands and more...

Q

Wishlist

Cart

Hi, Naveen

Categories

Home

Deals

Trending

Best Sellers

New Arrivals

Track Order

Home > Shopping Cart

Shopping Cart

4 items



Minimalist Ceramic Vase Set (3 Pcs)

Homely

-

1

+

₹2,199



Samsung Galaxy S24 (8GB/256GB)

Samsung

-

1

+

₹59,999



ASUS Vivobook 15 (Ryzen 5/8GB/512GB)

ASUS

-

1

+

₹45,990



Nike Air Max Shoes

Nike

-

1

+

₹4,999

Continue Shopping

Order Summary

Enter coupon code

Apply

Subtotal (4 items)

₹1,13,187

Shipping

FREE

Total

₹1,13,187

Proceed to Checkout

Secure

Fast Delivery

Easy Return

Product Details

S

ShopEZ

Shop Smarter. Live Better.

Search for products, brands and more...

Q

Wishlist

Cart

Hi, Naveen

Categories

Home

Deals

Trending

Best Sellers

New Arrivals

Track Order

Home > Home & Kitchen > Minimalist Ceramic Vase Set (3 Pcs)

27% OFF



Minimalist Ceramic Vase Set (3 Pcs)

★★★★★ (0)

178 sold

₹2,199

₹2,999

27% OFF

Inclusive of all taxes

✓ In Stock

Quantity:

-

1

+

Add to Cart

Buy Now

Free Delivery

On orders above ₹499

30 Days Returns

Easy return policy

Secure Payment

100% secure payments

Details

Price History

Reviews

Product Details

Set of 3 matte ceramic vases in varying heights. Perfect for dried flowers, pampas grass, or as standalone decor.

Checkout

S

ShopEZ

Shop Smarter. Live Better.

Q

Search for products, brands and more...

Q

Wishlist

Cart

NHi, Naveen

Categories

Home

Deals

Trending

Best Sellers

New Arrivals

Track Order

Checkout

Address

Delivery

Payment

Review

1

Shipping Address

Full Name *

Naveen kumar

Phone *

4841689489

Address Line 1 *

123, Green Park

Address Line 2

Apartment, floor, landmark (optional)

City *

New Delhi

State *

Delhi

Postal Code *

110016

Country

India

Or choose a saved address:

Home — Naveen kumar, Chinahamsa, Palasa

Continue to Delivery >

2

Delivery Options

3

Payment Method

4

Review Order

Order Summary

Minimalist Ceramic Vase Set...

Qty: 1

₹2,199

Samsung Galaxy S24 (8G...

Qty: 1

₹59,999

ASUS Vivobook 15 (Ryzen ...

Qty: 2

₹91,980

Nike Air Max Shoes

Qty: 1

₹4,999

Items (4)

₹1,59,177

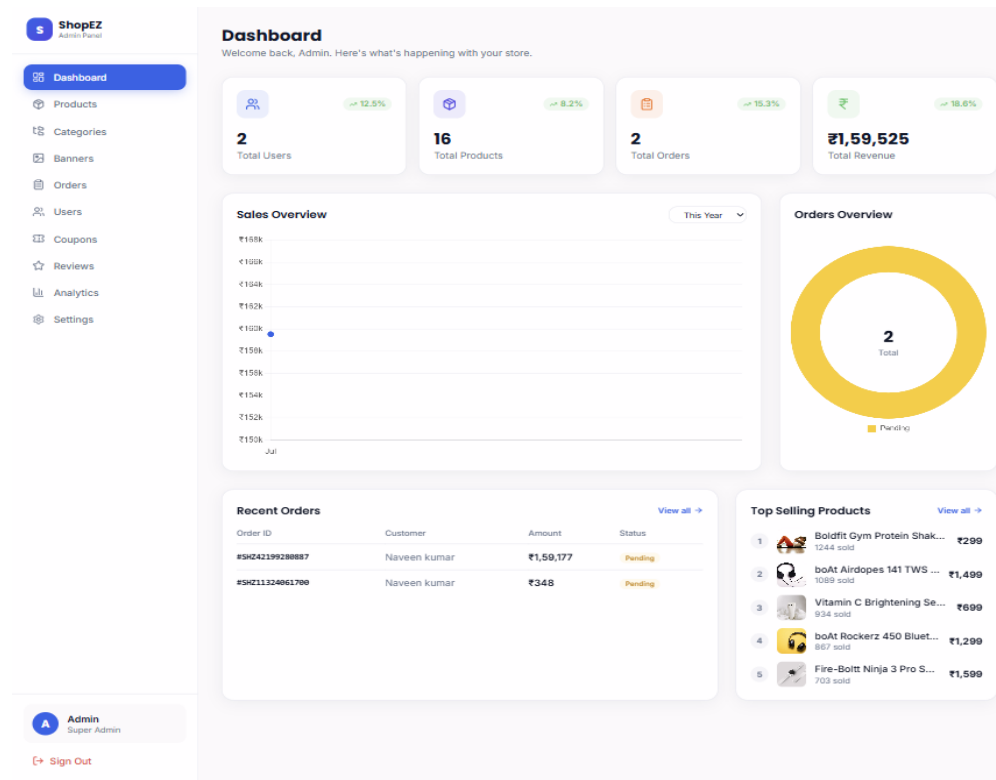
Shipping

FREE

Total

₹1,59,177

Admin Dashboard



8. Advantages & Disadvantages

Advantages

- Fully responsive interface usable across devices.
- Secure JWT-based authentication and hashed password storage.
- Clear separation of concerns via a three-tier MERN architecture.
- Centralized admin dashboard for managing products, users, and orders.
- RESTful APIs make the backend easy to extend or integrate with other clients.

Disadvantages

- Payment gateway integration is currently simulated, not live.
- No email notifications for order confirmation or password recovery yet.
- No product recommendation or review system in the current version.
- Performance under very high concurrent load has not been fully load-tested.
- Admin roles are currently basic, without granular permission levels.

9. Conclusion

SHOPEZ successfully demonstrates a complete MERN-stack e-commerce workflow, from product discovery through checkout, alongside an administrative dashboard for managing the storefront. The project gave the team practical experience across frontend development, REST API design, database modeling, and authentication, resulting in a functional, responsive, and reasonably secure shopping application suitable as a foundation for further development.

10. Future Scope

- Integrate secure online payment gateways such as Stripe, Razorpay, or PayPal.
- Implement email verification and password recovery functionality.
- Add product reviews and customer ratings.
- Develop a wishlist feature for saving favorite products.
- Introduce AI-based product recommendations based on user preferences.
- Implement real-time order tracking using WebSockets.
- Develop a multi-vendor marketplace with seller registration and management.
- Deploy the application using cloud platforms such as AWS, Azure, or Render.
- Add multilingual support and multiple currency options.
- Strengthen security with two-factor authentication (2FA).

11. Appendix

Source Code (if any)

Repository / Source:	https://github.com/Bhanusri-274/E-commerce-Application
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GitHub & Project Demo Link

GitHub Repository:	https://github.com/Bhanusri-274/E-commerce-Application
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Project Demo:	https://drive.google.com/file/d/1WYGA5G2TI2Y8IDUXmN9qU7v68w0Kuqfv/view?usp=drive_link
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